

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION

Federal State Autonomous Educational
Institution of Higher Education "Southern Federal University"
(SOUTHERN FEDERAL UNIVERSITY)

COURSE SYLLABUS FOR
Databases and DBMS (Базы данных и СУБД)

Подписано электронной подписью:

Г.Е. Веселов, Директор института компьютерных
технологий и информационной безопасности

Сертификат № 0643656F00F3B21B9D460D25A94BE5A817
действителен с 5 июня 2025 г. по 5 июня 2026 г.

Compiler(s) of the course syllabus:

S. A. Kuchеров, Candidate of Technical Sciences, Associate Professor of the Department of System Analysis and Telecommunications

A. I. Kostyuk, Candidate of Technical Sciences, Associate Professor, Associate Professor of the Department of Computer Engineering

E.M. Pilipushko, Senior Lecturer of the Department of Mathematical Support and Computer Application

The program was approved at the meeting of the International Council of the Institute of Computer Technologies and Information Security

June 19, 2025, Minutes No 4

TABLE OF CONTENTS

I. Goals and objectives of mastering the discipline	4
II. Place of the discipline in the structure of the educational program	4
III. Requirements for the results of mastering the discipline.....	6
IV. Content and structure of the discipline	21
4.1. The content of the discipline, structured by topics (educational meetings) for the 4th semester of full-time study	21
4.2. The content of the discipline, structured by topics (educational meetings) for the 11th trimester of part-time education.....	28
4.3. The content of the discipline, structured by topics (educational meetings) for the 12th trimester of part-time education.....	28
4.4. Plan of extracurricular independent work	33
4.5. Plan of extracurricular independent work for part-time education	38
4.6. Classes implemented in the form of practical training.....	43
V. Educational and methodological support of the discipline	44
5.1. Reference citations	44
5.2. Further reading	44
5.3. List of Internet resources.....	45
VI. Material and technical support of the discipline	45
VII. Methodical Guidelines for Students on Mastering the Discipline.....	45
VIII. Fund of Assessment Tools	46
8.1. Passport of the Appraisal Tools Fund	46
8.2. Computer Testing No1, No2	48
8.3. Laboratory Job No1	72
8.4. Laboratory Job No2.....	73
8.5. Laboratory Job No3.....	74
8.6. Laboratory Job No4.....	76
8.7. Laboratory Job No5.....	77
8.8. Laboratory Job No6.....	79
8.9. Laboratory Job No7.....	79
8.10. Laboratory Job No8.....	80
8.11. Exam Task.....	81

I. GOALS AND OBJECTIVES OF MASTERING THE DISCIPLINE

Objectives of mastering the discipline:

– study by the student of methods of database design, modern database management systems and obtaining practical skills in working with modern databases.

Objectives of mastering the discipline:

– gaining experience in one of the modern DBMS and obtaining knowledge, skills and abilities in the field of database design and administration, database software development.

II. THE PLACE OF THE DISCIPLINE IN THE STRUCTURE OF THE EDUCATIONAL PROGRAM

The discipline belongs to the compulsory part of the educational program and is included in the module of compulsory professional disciplines.

To study this discipline, it is necessary to have knowledge, skills and abilities formed by the previous elements of the educational program:

Name disciplines (modules), practices	Required knowledge, skills, and abilities
Algorithmization and programming	Knowledge: – Features of Programming in Modern Tool Environments – Data types and structures – principles of building programs in high-level languages – Syntax and semantics of high-level programming languages
	Skills: – algorithmization of actions to solve the problem – Algorithm optimization
	Skills: – Writing programs in high-level languages
Discrete Mathematics	Knowledge: – Boolean Algebra – Set theory – Operations on sets
	Skills: – Solving problems in set theory
	Skills: – performing operations on relationships
Object-oriented programming	Knowledge: – typical algorithms and data structures; – the dependence of the complexity of the software implementation of algorithms on the complexity of the data structure; – ways of presenting and encoding various types of information.
	Skills: – draw up elements of program documentation in accordance with the requirements of state and local regulatory documents; – apply standard algorithms and data structures to solve the problem; – represent numerical data in various codes, perform operations on them; – create program code, implement individual fragments of complex programs to solve professional problems

Name disciplines (modules), practices	Required knowledge, skills, and abilities
	Skills: – preparation of reports on the created algorithms and programs – development of algorithms for solving problems of professional activity, assessment of the complexity of the algorithm – use of a modern programming environment and the use of software libraries for program development; – Using a modern development environment to debug program code.

Knowledge, skills and abilities formed by this discipline will be required when mastering the following elements of the educational program:

- Creative project;
- Preparation for the defense procedure and defense of the final qualification work.

III. REQUIREMENTS FOR THE RESULTS OF MASTERING THE DISCIPLINE

The development of the discipline is aimed at the formation of the following competencies in accordance with the educational standard and educational program:

List of planned learning outcomes in the discipline correlated with indicators of competence achievement

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
09.03.03	CPC-2 Able to solve the problems of professional activity using modern information and communication technologies, hardware and software, including domestic production, and taking into account the basic requirements of information security and professional ethics	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	Knowledge: – modern technologies for structuring and storing data – Modern database management systems
			Skills: – Designing databases to store information – writing SQL queries, stored procedures, triggers – Database Object Development
			Skills: – working with the SQL language – Work in modern relational database management systems
	CPC-3 Able to participate in the development of regulatory, technical and reporting documentation, present the results of professional activities using standards, norms and rules	CPC-3.1 Draws up regulatory, technical and reporting documentation, taking into account standards, norms and rules	Knowledge: – graphical notations used in database design – SQL Query Code Notation
			Skills: – Documenting the results of the database design process – SQL Code Documentation
			Skills: – development of documentation for databases and database objects
	CPC-4. Able to develop algorithms and programs suitable for practical application in the field of professional activity	CPC-4.1 Selects data structures and develops algorithms for solving problems of professional activity	Knowledge: – Principles of structuring data in databases – data models, main advantages and disadvantages, and applications

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
			Skills: – Structure data according to domain specifics and software requirements – Algorithmization of procedures for manipulating data in databases
			Skills: – development of SQL queries – development of procedures and functions in PL/pgSQL
		CPC-4.2 Develops a software implementation of the algorithm	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL
			Skills: – working with the SQL interpreter in the PostgreSQL RDBMS
		CPC-4.3. Performs debugging and testing of programs	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing – principles and criteria for the optimality of SQL queries
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL – Debugging and optimizing SQL queries

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	CPC-5. Able to install, administer and configure software and hardware for information and automated systems	CPC-5.1. Installs and administers software based on modern standards of information interaction of systems	Skills: – Testing and debugging SQL queries using PostgreSQL
			Skills: – management of the process of installation and further configuration of the PostgreSQL DBMS in accordance with the specifics of use
		CPC-5.2. Performs parametric configuration of software for information and automated systems	Skills: – PostgreSQL DBMS installation in accordance with the specifics of use – PostgreSQL DBMS administration
			Knowledge: – settings and options available to the PostgreSQL DBMS administrator
			Skills: – Configuring PostgreSQL DBMS for Use Tasks
			Skills: – creating users in the PostgreSQL DBMS – security policy management in PostgreSQL DBMS – database management in PostgreSQL DBMS
09.03.01	CPC-2 Able to solve the problems of professional activity using modern information and communication technologies, hardware and software, including domestic production, and taking into account the basic requirements of information security and professional ethics	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	Knowledge: – modern technologies for structuring and storing data – Modern database management systems
			Skills: – Designing databases to store information – writing SQL queries, stored procedures, triggers – Database Object Development
			Skills: – working with the SQL language – Work in modern relational database management systems

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	CPC-3 Able to participate in the development of regulatory, technical and reporting documentation, present the results of professional activities using standards, norms and rules	CPC-3.1 Draws up regulatory, technical and reporting documentation, taking into account standards, norms and rules	Knowledge: – graphical notations used in database design – SQL Query Code Notation
			Skills: – Documenting the results of the database design process – SQL Code Documentation
			Skills: – development of documentation for databases and database objects
	CPC-4. Able to develop algorithms and programs suitable for practical application in the field of professional activity	CPC-4.1 Selects data structures and develops algorithms for solving problems of professional activity	Knowledge: – Principles of structuring data in databases – data models, main advantages and disadvantages, and applications
			Skills: – Structure data according to domain specifics and software requirements – Algorithmization of procedures for manipulating data in databases
			Skills: – development of SQL queries – development of procedures and functions in PL/pgSQL
		CPC-4.2 Develops a software implementation of the algorithm	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
			Skills: – working with the SQL interpreter in the PostgreSQL RDBMS
		CPC-4.3. Performs debugging and testing of programs	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing – principles and criteria for the optimality of SQL queries
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL – Debugging and optimizing SQL queries
			Skills: – Testing and debugging SQL queries using PostgreSQL
	CPC-5. Able to install, administer and configure software and hardware for information and automated systems	CPC-5.1. Installs and administers software based on modern standards of information interaction of systems	Skills: – management of the process of installation and further configuration of the PostgreSQL DBMS in accordance with the specifics of use
			Skills: – PostgreSQL DBMS installation in accordance with the specifics of use – PostgreSQL DBMS administration
		CPC-5.2. Performs parametric configuration of software for information and automated systems	Knowledge: – settings and options available to the PostgreSQL DBMS administrator Skills: – Configuring PostgreSQL DBMS for Use Tasks

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	CPC-7. Able to master the methods of using software to solve practical problems	GPK-7.1. Assesses the possibilities of using software to solve practical problems	Skills: – creating users in the PostgreSQL DBMS – security policy management in PostgreSQL DBMS – database management in PostgreSQL DBMS
			Knowledge: – modern technologies for structuring and storing data – Modern database management systems – Principles of structuring data in databases
			Skills: – Choosing the Right Database Architecture
		CPC-7.2. Analyzes technical documentation on the use of the software	Skills: – Evaluating Alternatives in Database Schema Development
			Knowledge: – SQL Query Code Notation
			Skills: – Work with documentation for database management systems – Work with documentation for SQL functions and procedures
		CPC-7.3. Selects and uses the necessary functions of software to solve the assigned tasks	Skills: – working with PostgreSQL documentation
			Knowledge: – Modern database management systems – settings and options available to the PostgreSQL DBMS administrator
			Skills: – management of the process of installation and further configuration of the PostgreSQL DBMS in accordance with the specifics of use

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
			Skills: – PostgreSQL DBMS installation in accordance with the specifics of use – PostgreSQL DBMS administration
09.03.02	CPC-2 Able to solve the problems of professional activity using modern information and communication technologies, hardware and software, including domestic production, and taking into account the basic requirements of information security and professional ethics	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	Knowledge: – modern technologies for structuring and storing data – Modern database management systems
			Skills: – Designing databases to store information – writing SQL queries, stored procedures, triggers – Database Object Development
			Skills: – working with the SQL language – Work in modern relational database management systems
	CPC-3 Able to participate in the development of regulatory, technical and reporting documentation, present the results of professional activities using standards, norms and rules	CPC-3.1 Draws up regulatory, technical and reporting documentation, taking into account standards, norms and rules	Knowledge: – graphical notations used in database design – SQL Query Code Notation
			Skills: – Documenting the results of the database design process – SQL Code Documentation
			Skills: – development of documentation for databases and database objects
	CPC-4. Able to develop algorithms and programs suitable for practical application in the field of professional activity	CPC-4.1 Selects data structures and develops algorithms for solving problems of professional activity	Knowledge: – Principles of structuring data in databases – data models, main advantages and disadvantages, and applications

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
			Skills: – Structure data according to domain specifics and software requirements – Algorithmization of procedures for manipulating data in databases
			Skills: – development of SQL queries – development of procedures and functions in PL/pgSQL
		CPC-4.2 Develops a software implementation of the algorithm	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL
			Skills: – working with the SQL interpreter in the PostgreSQL RDBMS
		CPC-4.3. Performs debugging and testing of programs	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing – principles and criteria for the optimality of SQL queries
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL – Debugging and optimizing SQL queries

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	CPC-5. Able to install software and hardware for information and automation systems	CPC-5.1. Installs software based on modern standards of information interaction of systems	Skills: – Testing and debugging SQL queries using PostgreSQL
			Skills: – management of the process of installation and further configuration of the PostgreSQL DBMS in accordance with the specifics of use
		CPC-5.2. Performs software configuration of information and automated systems	Skills: – PostgreSQL DBMS installation in accordance with the specifics of use – PostgreSQL DBMS administration
			Knowledge: – settings and options available to the PostgreSQL DBMS administrator
			Skills: – Configuring PostgreSQL DBMS for Use Tasks
			Skills: – creating users in the PostgreSQL DBMS – security policy management in PostgreSQL DBMS – database management in PostgreSQL DBMS
09.03.04	CPC-2 Able to solve the problems of professional activity using modern information and communication technologies, hardware and software, including domestic production, and taking into account the basic requirements of information security and professional ethics	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	Knowledge: – modern technologies for structuring and storing data – Modern database management systems
			Skills: – Designing databases to store information – writing SQL queries, stored procedures, triggers – Database Object Development
			Skills: – working with the SQL language – Work in modern relational database management systems

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	CPC-3 Able to participate in the development of regulatory, technical and reporting documentation, present the results of professional activities using standards, norms and rules	CPC-3.1 Draws up regulatory, technical and reporting documentation, taking into account standards, norms and rules	Knowledge: – graphical notations used in database design – SQL Query Code Notation
			Skills: – Documenting the results of the database design process – SQL Code Documentation
			Skills: – development of documentation for databases and database objects
	CPC-4. Able to develop algorithms and programs suitable for practical application in the field of professional activity	CPC-4.1 Selects data structures and develops algorithms for solving problems of professional activity	Knowledge: – Principles of structuring data in databases – data models, main advantages and disadvantages, and applications
			Skills: – Structure data according to domain specifics and software requirements – Algorithmization of procedures for manipulating data in databases
			Skills: – development of SQL queries – development of procedures and functions in the Transact-SQL language
		CPC-4.2 Develops a software implementation of the algorithm	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
			Skills: – working with the SQL interpreter in the Microsoft SQL-Server RDBMS
		CPC-4.3. Performs debugging and testing of programs	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing – principles and criteria for the optimality of SQL queries
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL – Debugging and optimizing SQL queries
			Skills: – Testing and debugging SQL queries using Microsoft SQL-Server
	CPC-5. Able to install and maintain software for information systems and databases, including domestic production	CPC-5.1. Installs and administers software based on modern standards of information interaction of systems	Skills: – management of the process of installation and further configuration of the Microsoft SQL Server DBMS in accordance with the specifics of use
			Skills: – Installation of Microsoft SQL Server DBMS in accordance with the specifics of use – Microsoft SQL Server DBMS Administration
		CPC-5.2. Performs parametric configuration of software for information and automated systems	Knowledge: – settings and settings available to Microsoft SQL Server DBMS Administrator Skills: – Configuration of Microsoft SQL Server DBMS for the tasks of use

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	CPC-6. Is able to search, store, process and analyze information from various sources and databases, present it in the required format using information, computer and network technologies	CPC-6.1. Performs information search using information, computer and network technologies	Skills: – creating users in the Microsoft SQL Server DBMS – Managing Security Policy in Microsoft SQL Server – Database management in Microsoft SQL Server
			Knowledge: – Principles of data retrieval from relational databases
			Skills: – development of algorithms for data selection in SQL
		GPK-6.2. Chooses the method of presentation, organizes the storage and processing of information using computer technology	Skills: – development of SQL queries
			Knowledge: – Principles of structuring data in databases – data models, main advantages and disadvantages, and applications
			Skills: – Structure data according to domain specifics and software requirements – Algorithmization of procedures for manipulating data in databases
10.03.01	CPC-2 Able to solve the problems of professional activity using modern information and communication technologies, hardware and software,	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	Skills: – development of SQL queries – development of procedures and functions in the Transact-SQL language
			Knowledge: – modern technologies for structuring and storing data – Modern database management systems

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
	including domestic production, and taking into account the basic requirements of information security and professional ethics		Skills: – Designing databases to store information – writing SQL queries, stored procedures, triggers – Database Object Development
			Skills: – working with the SQL language – Work in modern relational database management systems
	CPC-3 Able to participate in the development of regulatory, technical and reporting documentation, present the results of professional activities using standards, norms and rules	CPC-3.1 Draws up regulatory, technical and reporting documentation, taking into account standards, norms and rules	Knowledge: – graphical notations used in database design – SQL Query Code Notation
			Skills: – Documenting the results of the database design process – SQL Code Documentation
			Skills: – development of documentation for databases and database objects
	CPC-4. Able to develop algorithms and programs suitable for practical application in the field of professional activity	CPC-4.1 Selects data structures and develops algorithms for solving problems of professional activity	Knowledge: – Principles of structuring data in databases – data models, main advantages and disadvantages, and applications
			Skills: – Structure data according to domain specifics and software requirements – Algorithmization of procedures for manipulating data in databases
			Skills: – development of SQL queries – development of procedures and functions in the Transact-SQL language

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
		CPC-4.2 Develops a software implementation of the algorithm	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL
			Skills: – working with the SQL interpreter in the Microsoft SQL-Server RDBMS
		CPC-4.3. Performs debugging and testing of programs	Knowledge: – Principles of data retrieval from relational databases – Features of the use of unary and binary operations in the implementation of data processing – principles and criteria for the optimality of SQL queries
			Skills: – development of algorithms for selecting, inserting, deleting and modifying data in SQL – Debugging and optimizing SQL queries
			Skills: – Testing and debugging SQL queries using Microsoft SQL-Server
09.03.01 09.03.02 09.03.03 09.03.04	PC-6. BD-3 Able to organize data storage, choosing adequate technological solutions	PC-6.1 BD-3.1 Develops, debugs and tests application solutions with AI elements using various structured data storage technologies, evaluates the quality	Knowledge: – modern technologies for structuring and storing data – Modern database management systems

Code of the direction of training, specialty	Competence	Indicators of Competency Achievement	Learning outcomes
10.03.01			Skills: – Designing databases to store information – writing SQL queries, stored procedures, triggers – Database Object Development
			Skills: – working with the SQL language – Work in modern relational database management systems
		PC-6.2 BD-3.2 Develops, debugs and tests application solutions with AI elements using various technologies for storing unstructured data, assesses the quality	Knowledge: – Modern technologies for storing unstructured data – NoSQL database management systems
			Skills: – organization of storage of unstructured data – writing CRUD procedures for unstructured data – Database Object Development
			Skills: – work with NoSQL DBMS and built-in languages

IV. CONTENT AND STRUCTURE OF THE DISCIPLINE

The complexity of the discipline is 5 credits, 180 hours,
including 1 credit, 36 hours for the exam

Form of intermediate certification: exam

4.1. The content of the discipline, structured by topics (educational meetings) for the 4th semester of full-time study

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
Module 1. Basics of Databases										
1	Topic 1. Introduction to the course Basic concepts, data models, DBMS	2	–	–	–	2	Informational lecture	–	–	–
2	Topic 2. Data models Relational Data Model and Relational Algebra	2	–	–	–	2	Informational lecture	–	–	–
3	Topic 3. Database Design Database Design	2	–	–	–	2	Informational lecture	–	–	–
4	Topic 3. Database Design Formation of technical specifications and database design	–	–	2	–	2	Reproductive laboratory work	–	–	–
5	Topic 3. Database Design Normalization of relations	2	–	–	–	2	Informational lecture	–	–	–
6	Topic 3. Database Design Formation of technical specifications and database design	–	–	2	–	4	Reproductive laboratory work	Laboratory work No1	Report protection	7
7	Topic 4. Database development Installing PostgreSQL, Data Types	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
8	Topic 4. Database development Basic Postgres objects, DDL commands	2	–	–	–	2	Informational lecture	–	–	–
9	Topic 4. Database development Physical design and development and initial filling of the database	–	–	2	–	2	Reproductive laboratory work	–	–	–
10	Topic 4. Database development Create, modify, and delete Postgres objects, DDL commands	2	–	–	–	2	Informational lecture	–	–	–
11	Topic 4. Database development Physical design and development and initial filling of the database	–	–	2	–	4	Reproductive laboratory work	Laboratory Work No2	Report protection	7
12	Topic 5. Managing Database Content DML. Database Selection: SELECT, JOIN, and Set Operations	2	–	–	–	2	Informational lecture	–	–	–
13	Topic 5. Managing Database Content Complex data sampling queries	–	–	2	–	2	Reproductive laboratory work	–	–	–
14	Topic 5. Managing Database Content DML. Operations on sets. Aggregation. Window functions.	2	–	–	–	2	Informational lecture	–	–	–
15	Topic 5. Managing Database Content Complex data sampling queries	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No3	Report protection	5

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
16	Topic 5. Managing Database Content Data Modification	2	–	–	–	2	Informational lecture	Computer Testing No1	Test	10
Module 2. PostgreSQL development										
17	Topic 6. Programming in the PostgreSQL environment Server-side programming	2	–	–	–	2	Informational lecture	–	–	–
18	Topic 6. Programming in the PostgreSQL environment Implementing business logic in a database	–	–	2	–	2	Reproductive laboratory work	–	–	–
19	Topic 6. Programming in the PostgreSQL environment Built-in features. Triggers. Full-text search in the database. Transactions	2	–	–	–	2	Informational lecture	–	–	–
20	Topic 6. Programming in the PostgreSQL environment Implementing business logic in a database	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No4	Report protection	5
21	Topic 7. Managing Databases in a PostgreSQL Environment Types of indexes in PostgreSQL.	2	–	–	–	2	Informational lecture	–	–	–
22	Topic 7. Managing Databases in a PostgreSQL Environment Applied tasks of using databases	–	–	2	–	2	Reproductive laboratory work	–	–	–
23	Topic 7. Managing Databases in a PostgreSQL Environment Applied tasks of using databases	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No5	Report protection	5

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
24	Topic 7. Managing Databases in a PostgreSQL Environment Dynamic commands, arrays, error handling. Cursors and database backups. Access control in PostgreSQL. Encoding information in PostgreSQL	2	–	–	–	2	Informational lecture	–	–	–
25	Topic 7. Managing Databases in a PostgreSQL Environment Working with arrays, JSON, and encoding information	–	–	4	–	2	Reproductive laboratory work	Laboratory Work No6	Report protection	5
Module 3. Data Storage in Artificial Intelligence Systems										
26	Topic 8. Data Storage in Artificial Intelligence An introduction to unstructured data and its role in AI. The concept and features of unstructured data (text, images, audio, video, event logs, sensory data). Differences between structured, semi-structured, and unstructured data. Artificial intelligence tasks that work with unstructured data (NLP, CV, Speech, Multimodal AI). Storage requirements for AI (scalability, availability, schema flexibility). Overview of popular platforms: HDFS, MongoDB, Elasticsearch, MinIO, AWS S3, PostgreSQL JSONB, Vector Databases (FAISS, Milvus, Weaviate).	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
27	Topic 8. Data Storage in Artificial Intelligence Storing and processing unstructured text data	—	—	4	—	4	Reproductive laboratory work	—	—	—
28	Topic 8. Data Storage in Artificial Intelligence Document repositories and semi-structured data. Document-oriented database model. Storage formats: JSON, BSON, Avro, Parquet. Principles of indexing and searching on unstructured data. MongoDB, CouchDB, Elasticsearch. The use of technologies in the tasks of text analysis and building NLP models. Organize a corpus of texts to train a classification/sentiment model.	2	—	—	—	2	Informational lecture	—	—	—
29	Topic 8. Data Storage in Artificial Intelligence Storing and processing unstructured text data	—	—	2	—	2	Reproductive laboratory work	Laboratory Work No7	Report protection	5
30	Topic 8. Data Storage in Artificial Intelligence Storage and media management for AI. Features of storing large multimedia data sets. Blob storage and object systems (MinIO, S3, Azure Blob Storage). Content metadata and descriptors. Integration with ML platforms (TensorFlow, PyTorch	2	—	—	—	2	Informational lecture	—	—	—

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
	Datasets, DVC, MLflow). Version control of datasets and reproducibility of experiments. Organize an image repository for an object recognition task.									
31	Topic 8. Data Storage in Artificial Intelligence Vector databases and intelligent search. Embeddings. Principles of building vector stores (FAISS, Milvus, Pinecone, Weaviate). Nearest neighbor search (ANN) algorithms. The architecture of RAG (Retrieval-Augmented Generation) systems and their relationship to LLM. Storing and searching embeddings of texts or images	2	–	–	–	4	Informational lecture	Computer Testing No2	Test	10
32	Topic 8. Data Storage in Artificial Intelligence Vector embeddings storage and intelligent search	–	–	4	–	4	Reproductive laboratory work			
33	Topic 8. Data Storage in Artificial Intelligence Vector embeddings storage and intelligent search	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No8	Report protection	5
Intermediate certification <i>(for disciplines with an exam)</i>		–	–	–	–	36		Exam Questions and Tickets for Semester 4	<i>Exam</i>	40
Total Hours		36		36		108		–		100

4.2. The content of the discipline, structured by topics (educational meetings) for the 11th trimester of part-time education

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
Module 1. Basics of Databases										
1	Topic 1. Introduction to the course Basic concepts, data models, DBMS	2	–	–	–	34	Informational lecture	–	–	–
Total Hours		2		–		34		–		–

4.3. The content of the discipline, structured by topics (educational meetings) for the 12th trimester of part-time education

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
Module 1. Basics of Databases										
1	Topic 2. Data models Relational Data Model and Relational Algebra	–	–	–	–	6	Self-study	–	–	–
2	Topic 3. Database Design Database Design	2	–	–	–	2	Informational lecture	–	–	–
3	Topic 3. Database Design Normalization of relations	2	–	–	–	2	Informational lecture	–	–	–
4	Topic 3. Database Design Formation of technical specifications and database design	–	–	2	–	6	Reproductive laboratory work	Laboratory work No1	Report protection	7
5	Topic 4. Database development Installing PostgreSQL, Data Types	–	–	–	–	6	Self-study	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
6	Topic 4. Database development Basic Postgres objects, DDL commands	–	–	–	–	6	Self-study	–	–	–
7	Topic 4. Database development Create, modify, and delete Postgres objects, DDL commands	2	–	–	–	2	Informational lecture	–	–	–
8	Topic 4. Database development Physical design and development and initial filling of the database	–	–	2	–	6	Reproductive laboratory work	Laboratory Work No2	Report protection	7
9	Topic 5. Managing Database Content DML. Database Selection: SELECT, JOIN, and Set Operations	2	–	–	–	2	Informational lecture	–	–	–
10	Topic 5. Managing Database Content DML. Operations on sets. Aggregation. Window functions.	–	–	–	–	6	Self-study	–	–	–
11	Topic 5. Managing Database Content Complex data sampling queries	–	–	2	–	6	Reproductive laboratory work	Laboratory Work No3	Report protection	7
12	Topic 5. Managing Database Content Data Modification	–	–	–	–	6	Self-study	Computer Testing No1	Test	10
Module 2. PostgreSQL development										
13	Topic 6. Programming in the PostgreSQL environment Server-side programming	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
14	Topic 6. Programming in the PostgreSQL environment Built-in features. Triggers. Full-text search in the database	2	–	–	–	2	Informational lecture	–	–	–
15	Topic 6. Programming in the PostgreSQL environment Implementing business logic in a database	–	–	2	–	6	Reproductive laboratory work	Laboratory Work No4	Report protection	7
16	Topic 7. Managing Databases in a PostgreSQL Environment Transactions	–	–	–	–	6	Self-study	–	–	–
17	Topic 7. Managing Databases in a PostgreSQL Environment Types of indexes in PostgreSQL	–	–	–	–	6	Self-study	–	–	–
18	Topic 7. Managing Databases in a PostgreSQL Environment Applied tasks of using databases	–	–	2	–	6	Reproductive laboratory work	Laboratory Work No5	Report protection	7
19	Topic 7. Managing Databases in a PostgreSQL Environment Dynamic commands, arrays, error handling. Cursors and database backups. Access control in PostgreSQL. Encoding information in PostgreSQL	–	–	–	–	6	Self-study			
20	Topic 7. Managing Databases in a PostgreSQL Environment Working with arrays, JSON, and encoding information	–	–	2	–	6	Reproductive laboratory work	Laboratory Work No6	Report protection	5
Module 3. Data Storage in Artificial Intelligence Systems										
21	Topic 8. Data Storage in Artificial Intelligence	2	–	–	–	5	Informational lecture	Computer Testing No2	Test	10

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independen t work				
		Lectures	Practical	Laboratory	Consulting					
	An introduction to unstructured data and its role in AI. The concept and features of unstructured data (text, images, audio, video, event logs, sensory data). Differences between structured, semi-structured, and unstructured data. Artificial intelligence tasks that work with unstructured data (NLP, CV, Speech, Multimodal AI). Storage requirements for AI (scalability, availability, schema flexibility). Overview of popular platforms: HDFS, MongoDB, Elasticsearch, MinIO, AWS S3, PostgreSQL JSONB, Vector Databases (FAISS, Milvus, Weaviate).									
22	Topic 8. Data Storage in Artificial Intelligence Document repositories and semi-structured data. Document-oriented database model. Storage formats: JSON, BSON, Avro, Parquet. Principles of indexing and searching on unstructured data. MongoDB, CouchDB, Elasticsearch. The use of technologies in the tasks of text analysis and building NLP models. Organize a corpus of texts to train a classification/sentiment model.	—	—	—	—	2	Self-study	—	—	—

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
23	Topic 8. Data Storage in Artificial Intelligence Storage and media management for AI. Features of storing large multimedia data sets. Blob storage and object systems (MinIO, S3, Azure Blob Storage). Content metadata and descriptors. Integration with ML platforms (TensorFlow, PyTorch Datasets, DVC, MLflow). Version control of datasets and reproducibility of experiments. Organize an image repository for an object recognition task.	–	–	–	–	2	Self-study	–	–	–
24	Topic 8. Data Storage in Artificial Intelligence Vector databases and intelligent search. Embeddings. Principles of building vector stores (FAISS, Milvus, Pinecone, Weaviate). Nearest neighbor search (ANN) algorithms. The architecture of RAG (Retrieval-Augmented Generation) systems and their relationship to LLM. Storing and searching embeddings of texts or images	–	–	–	–	4	Self-study			
Intermediate certification <i>(for disciplines with an exam)</i>		–	–	–	–	9		Exam Questions and Tickets for Semester 4	<i>Exam</i>	40
Total Hours		14	–	12	–	118		–		100

4.4. Plan of extracurricular independent work

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Basics of Databases						
1	Topic 1. Introduction to the course Basic concepts, data models, DBMS	4	1–9	– study and repetition of the material of lectures; – Preparation for testing	2	[1]–[3]; [8]–[14]
2	Topic 2. Data models Relational Data Model and Relational Algebra	4	1–9	– study and repetition of the material of lectures; – preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[1]–[3]; [8]–[14]
3	Topic 3. Database Design Database Design	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
4	Topic 3. Database Design Formation of technical specifications and database design	4	1–9	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[1]–[3]; [8]–[14]
5	Topic 3. Database Design Normalization of relations	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
6	Topic 3. Database Design Formation of technical specifications and database design	4	1–9	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[1]–[3]; [8]–[14]
7	Topic 4. Database development Installing PostgreSQL, Data Types	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
8	Topic 4. Database development Basic Postgres objects, DDL commands	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
9	Topic 4. Database development Physical design and development and initial filling of the database	4	1–9	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[1]–[3]; [8]–[14]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
10	Topic 4. Database development Create, modify, and delete Postgres objects, DDL commands	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
11	Topic 4. Database development Physical design and development and initial filling of the database	4	1–9	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[1]–[3]; [8]–[14]
12	Topic 5. Managing Database Content DML. Database Selection: SELECT, JOIN, and Set Operations	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
13	Topic 5. Managing Database Content Complex data sampling queries	4	1–9	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[1]–[3]; [8]–[14]
14	Topic 5. Managing Database Content DML. Operations on sets. Aggregation. Window functions.	4	1–9	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
15	Topic 5. Managing Database Content Complex data sampling queries	4	1–9	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[1]–[3]; [8]–[14]
16	Topic 5. Managing Database Content Data Modification	4	1–9	– study and repetition of the material of lectures; – Preparation for testing	2	[1]–[3]; [8]–[14]
Module 2. PostgreSQL development						
17	Topic 6. Programming in the PostgreSQL environment Server-side programming	4	10-14	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
18	Topic 6. Programming in the PostgreSQL environment Implementing business logic in a database	4	10-14	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[1]–[3]; [8]–[14]

[illegible]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
26	Topic 8. Data Storage in Artificial Intelligence An introduction to unstructured data and its role in AI. The concept and features of unstructured data (text, images, audio, video, event logs, sensory data). Differences between structured, semi-structured, and unstructured data. Artificial intelligence tasks that work with unstructured data (NLP, CV, Speech, Multimodal AI). Storage requirements for AI (scalability, availability, schema flexibility). Overview of popular platforms: HDFS, MongoDB, Elasticsearch, MinIO, AWS S3, PostgreSQL JSONB, Vector Databases (FAISS, Milvus, Weaviate).	4	15-18	– study and repetition of the material of lectures;	2	[4]–[7]
27	Topic 8. Data Storage in Artificial Intelligence Storing and processing unstructured text data	4	15-18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[4]–[7]
28	Topic 8. Data Storage in Artificial Intelligence Document repositories and semi-structured data. Document-oriented database model. Storage formats: JSON, BSON, Avro, Parquet. Principles of indexing and searching on unstructured data. MongoDB, CouchDB, Elasticsearch. The use of technologies in the tasks of text analysis and building NLP models. Organize a corpus of texts to train a classification/sentiment model.	4	15-18	– study and repetition of the material of lectures;	2	[4]–[7]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
29	Topic 8. Data Storage in Artificial Intelligence Storing and processing unstructured text data	4	15-18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[4]–[7]
30	Topic 8. Data Storage in Artificial Intelligence Storage and media management for AI. Features of storing large multimedia data sets. Blob storage and object systems (MinIO, S3, Azure Blob Storage). Content metadata and descriptors. Integration with ML platforms (TensorFlow, PyTorch Datasets, DVC, MLflow). Version control of datasets and reproducibility of experiments. Organize an image repository for an object recognition task.	4	15-18	– study and repetition of the material of lectures;	2	[4]–[7]
31	Topic 8. Data Storage in Artificial Intelligence Vector databases and intelligent search. Embeddings. Principles of building vector stores (FAISS, Milvus, Pinecone, Weaviate). Nearest neighbor search (ANN) algorithms. The architecture of RAG (Retrieval-Augmented Generation) systems and their relationship to LLM. Storing and searching embeddings of texts or images	4	15-18	– study and repetition of the material of lectures; – Preparation for testing	4	[4]–[7]
32	Topic 8. Data Storage in Artificial Intelligence Vector embeddings storage and intelligent search		15-18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[4]–[7]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
33	Topic 8. Data Storage in Artificial Intelligence Vector embeddings storage and intelligent search	4	15-18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[4]–[7]
Exam preparation					36	[1]–[14]
The total labor intensity of independent work in the discipline					108	–

4.5. Plan of extracurricular independent work for part-time education

No p/n	Topic of the lesson	Trimester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Basics of Databases						
1	Topic 1. Introduction to the course Basic concepts, data models, DBMS	11	1–18	– study and repetition of the material of lectures;	34	[1]–[3]; [8]–[14]
2	Topic 2. Data models Relational Data Model and Relational Algebra	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]
3	Topic 3. Database Design Database Design	12	1–18	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
4	Topic 3. Database Design Normalization of relations	12	1–18	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
5	Topic 3. Database Design Formation of technical specifications and database design	12	1–18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work;	6	[1]–[3]; [8]–[14]
6	Topic 4. Database development Installing PostgreSQL, Data Types	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]

No p/n	Topic of the lesson	Trimester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
7	Topic 4. Database development Basic Postgres objects, DDL commands	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]
8	Topic 4. Database development Create, modify, and delete Postgres objects, DDL commands	12	1–18	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
9	Topic 4. Database development Physical design and development and initial filling of the database	12	1–18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work;	6	[1]–[3]; [8]–[14]
10	Topic 5. Managing Database Content DML. Database Selection: SELECT, JOIN, and Set Operations	12	1–18	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
11	Topic 5. Managing Database Content DML. Operations on sets. Aggregation. Window functions.	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]
12	Topic 5. Managing Database Content Complex data sampling queries	12	1–18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work;	6	[1]–[3]; [8]–[14]
13	Topic 5. Managing Database Content Data Modification	12	1–18	– study and repetition of the material of lectures; – Preparation for testing	6	[1]–[3]; [8]–[14]
Module 2. PostgreSQL development						
14	Topic 6. Programming in the PostgreSQL environment Server-side programming	12	1–18	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
15	Topic 6. Programming in the PostgreSQL environment Built-in features. Triggers. Full-text search in the database	12	1–18	– study and repetition of the material of lectures;	2	[1]–[3]; [8]–[14]
16	Topic 6. Programming in the PostgreSQL environment Implementing business logic in a database	12	1–18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work;	6	[1]–[3]; [8]–[14]

No p/n	Topic of the lesson	Trimester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
17	Topic 7. Managing Databases in a PostgreSQL Environment Transactions	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]
18	Topic 7. Managing Databases in a PostgreSQL Environment Types of indexes in PostgreSQL	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]
19	Topic 7. Managing Databases in a PostgreSQL Environment Applied tasks of using databases	12	1–18	– preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work;	6	[1]–[3]; [8]–[14]
20	Topic 7. Managing Databases in a PostgreSQL Environment Dynamic commands, arrays, error handling. Cursors and database backups. Access control in PostgreSQL. Encoding information in PostgreSQL	12	1–18	– study and repetition of the material of lectures;	6	[1]–[3]; [8]–[14]
Module 3. Data Storage in Artificial Intelligence Systems						

No p/n	Topic of the lesson	Trimester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
21	Topic 8. Data Storage in Artificial Intelligence An introduction to unstructured data and its role in AI. The concept and features of unstructured data (text, images, audio, video, event logs, sensory data). Differences between structured, semi-structured, and unstructured data. Artificial intelligence tasks that work with unstructured data (NLP, CV, Speech, Multimodal AI). Storage requirements for AI (scalability, availability, schema flexibility). Overview of popular platforms: HDFS, MongoDB, Elasticsearch, MinIO, AWS S3, PostgreSQL JSONB, Vector Databases (FAISS, Milvus, Weaviate).	12	1–18	– study and repetition of the material of lectures; – Preparation for testing	4	[4]–[7]
22	Topic 8. Data Storage in Artificial Intelligence Document repositories and semi-structured data. Document-oriented database model. Storage formats: JSON, BSON, Avro, Parquet. Principles of indexing and searching on unstructured data. MongoDB, CouchDB, Elasticsearch. The use of technologies in the tasks of text analysis and building NLP models. Organize a corpus of texts to train a classification/sentiment model.	12	1–18	– study and repetition of the material of lectures;	2	[4]–[7]

No p/n	Topic of the lesson	Trimester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
23	Topic 8. Data Storage in Artificial Intelligence Storage and media management for AI. Features of storing large multimedia data sets. Blob storage and object systems (MinIO, S3, Azure Blob Storage). Content metadata and descriptors. Integration with ML platforms (TensorFlow, PyTorch Datasets, DVC, MLflow). Version control of datasets and reproducibility of experiments. Organize an image repository for an object recognition task.	12	1–18	– study and repetition of the material of lectures;	2	[4]–[7]
24	Topic 8. Data Storage in Artificial Intelligence Vector databases and intelligent search. Embeddings. Principles of building vector stores (FAISS, Milvus, Pinecone, Weaviate). Nearest neighbor search (ANN) algorithms. The architecture of RAG (Retrieval-Augmented Generation) systems and their relationship to LLM. Storing and searching embeddings of texts or images	12	1–18	– study and repetition of the material of lectures;	4	[4]–[7]
	Exam preparation				9	[1]–[14]
	The total labor intensity of independent work in the discipline				152	–

4.6. Classes implemented in the form of practical training

No p/n	Topic of the lesson	Type of educational work	Format of the lesson	Location of the class	Volume in hours
1	Topic 3. Database Design Formation of technical specifications and database design	Laboratory work	Reproductive laboratory work	SFedU	4
2	Topic 4. Database development Physical design and development and initial filling of the database	Laboratory work	Reproductive laboratory work	SFedU	4
3	Topic 5. Managing Database Content Complex data sampling queries	Laboratory work	Reproductive laboratory work	SFedU	4
4	Topic 6. Programming in the PostgreSQL environment Implementing business logic in a database	Laboratory work	Reproductive laboratory work	SFedU	4
5	Topic 7. Managing Databases in a PostgreSQL Environment Applied tasks of using databases	Laboratory work	Reproductive laboratory work	SFedU	4
6	Topic 7. Managing Databases in a PostgreSQL Environment Cursors and database backups	Laboratory work	Reproductive laboratory work	SFedU	4
7	Topic 8. Data Storage in Artificial Intelligence Storing and processing unstructured text data	Laboratory work	Reproductive laboratory work	SFedU	6
8	Topic 8. Data Storage in Artificial Intelligence Vector embeddings storage and intelligent search	Laboratory work	Reproductive laboratory work	SFedU	6
TOTAL:					36

V. EDUCATIONAL AND METHODOLOGICAL SUPPORT OF THE DISCIPLINE

5.1. References

1. Komarov V. I. Guide to Databases. Moscow: DMK Press, 2024. 520 p. <https://postgrespro.ru/education/books/dbguide>
2. Sidorova, N. P. Databases: a workshop on the design of relational databases: a textbook: [16+] / N. P. Sidorova; University of Technology, Institute of Engineering and Digital Technologies, Faculty of Infocommunication Systems and Technologies. – Moscow; Berlin: Direct-Media, 2020. – 93 p. : ill. – Mode of access: by subscription. Available at: <https://biblioclub.ru/index.php?page=book&id=575080> (accessed: 06.05.2025). – Bibliogr.: p. 85. – ISBN 978-5-4499-0799-8. – Text : electronic.
3. PostgreSQL. Professional SQL: textbook. posobiye / E. P. Morgunov; ed. by E. V. Rogov. — M. : DMK Press, 2025. 444 p. <https://postgrespro.ru/education/books/advancedsql>
4. James Jie Pan, Jianguo Wang, and Guoliang Li. 2024. Vector Database Management Techniques and Systems. In Companion of the 2024 International Conference on Management of Data (SIGMOD '24). Association for Computing Machinery, New York, NY, USA, 597–604. <https://doi.org/10.1145/3626246.3654691>
5. Ji Sun, Guoliang Li, James Pan, Jiang Wang, Yongqing Xie, Ruicheng Liu, and Wen Nie. 2025. GaussDB-Vector: A Large-Scale Persistent Real-Time Vector Database for LLM Applications. Proc. VLDB Endow. 18, 12 (August 2025), 4951–4963. <https://doi.org/10.14778/3750601.3750619>
6. Qian Xu, Feng Zhang, Chengxi Li, Lei Cao, Zheng Chen, Jidong Zhai, and Xiaoyong Du. 2025. HARMONY: A Scalable Distributed Vector Database for High-Throughput Approximate Nearest Neighbor Search. Proc. ACM Manag. Data 3, 4, Article 249 (September 2025), 28 pages. <https://doi.org/10.1145/3749167>
7. Alexander Thomasian, Chapter 9 - Structured, unstructured, and diverse databases, Editor(s): Alexander Thomasian, Storage Systems, Morgan Kaufmann, 2022, Pages 493-563, ISBN 9780323907965, <https://doi.org/10.1016/B978-0-32-390796-5.00018-8>.

5.2. Further reading

8. Luzanov P.V., Rogov E.V., Levshin I.V. Postgres. First acquaintance. Moscow: DMK Press, 2025. — 191 p. <https://postgrespro.ru/education/books/introbook>
9. Rogov E. V. PostgreSQL 17 from the inside. Moscow: DMK Press, 2025. — 668 p. <https://postgrespro.ru/education/books/internals>
10. Lesovsky A. V. Monitoring of PostgreSQL. Moscow, Bumba Publ., 2024. 247 p. <https://postgrespro.ru/education/books/monitoring>
11. Sidorova, N. P. Databases: a workshop on the design of relational databases: a textbook: [16+] / N. P. Sidorova; University of Technology, Institute of Engineering and Digital Technologies, Faculty of Infocommunication Systems and Technologies. – Moscow; Berlin: Direct-Media, 2020. – 93 p. : ill. – Mode of access: by subscription. Available at: <https://biblioclub.ru/index.php?page=book&id=575080> (accessed: 06.05.2025). – Bibliogr.: p. 85. – ISBN 978-5-4499-0799-8. – Text : electronic.
12. Gudov, A. M. Administration of database management systems : textbook : [16+] / A. M. Gudov, I. Yu. Kemerovo State University. – Kemerovo : Kemerovo State University, 2022. – 167 p. : ill. – Mode of access: by subscription. Available at: <https://biblioclub.ru/index.php?page=book&id=700656> (accessed: 06.05.2025). – ISBN 978-5-8353-2893-2. – Text : electronic.
13. Shilin, A. S. Perspektivnye metody proektirovaniya relational'nykh bazodnykh daz : uchebnoe posobie : [12+] / A. S. Shilin. – Moscow ; Berlin : Direct-Media, 2021. – 136 p. : ill., schemes, tabl. – Mode of access: by subscription. – URL: <https://biblioclub.ru/index.php?page=book&id=602240> (date of access: 06.05.2025). – Bibliogr. in the book – ISBN 978-5-4499-1890-1. – Text: electronic.
14. Fundamentals of building databases : textbook : [16+] / A. V. Averchenkov, R. A. Filippov, Y. A. Leonov [i dr.]. – Moscow; Berlin: Direct-Media, 2021. – 124 p. : ill., tabl. – Mode of access: by subscription. Available at: <https://biblioclub.ru/index.php?page=book&id=602227> (accessed: 06.05.2025). – Bibliogr. in the book. – ISBN 978-5-4499-2428-5. – Text : electronic.

5.3. List of Internet resources

- PostgreSQL 17.4 Documentation (<https://postgrespro.ru/docs/postgresql/current>);

VI. MATERIAL AND TECHNICAL SUPPORT OF THE DISCIPLINE

The following premises, equipment and software are used in the implementation of the discipline:

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
1	Lectures	Lecture Hall	– Multimedia projector; – Screen.
3	Laboratory Classes	Computer Lab	– Interactive whiteboard with a projector; – Personal computers (25 pcs.); – DBMS PostgreSQL or PostgresPRO; – Microsoft Windows or RED OS; – PgAdmin; – Microsoft Office or My Office.
4	Independent work	Self-study room	– Personal computer; – DBMS PostgreSQL or PostgresPRO; – Microsoft Windows or RED OS; – PgAdmin; – Microsoft Office or My Office.

VII. METHODOICAL INSTRUCTIONS FOR STUDENTS ON MASTERING THE DISCIPLINE

The educational process of learning in the discipline includes classroom classes (lectures and laboratory classes) and independent work. The final control in the discipline is an exam. The lecturer and teachers conducting laboratory classes control the attendance of all types of classroom classes.

Preparation for lectures. Acquaintance with the discipline takes place already at the first lecture. The student is required not only to pay attention, but also to independently draw up a note. When working with lecture notes, it is necessary to take into account the factor that some lectures give answers to specific questions of the topic, others only reveal the relationships between phenomena, helping the student to understand the deep processes of the development of the subject under study, both in history and at the present time.

Laboratory classes consist of analyzing and completing a common task together with the teacher and working on an independent task. The results of the work performed are shown to the teacher in the form of prepared reports. Based on the results of completing the tasks of each of the sections, the student receives points, the sum of which determines the final grade.

Independent work of students includes preparation for lectures, laboratory work and exams.

Ethics of using artificial intelligence. When mastering this course, it is allowed to use artificial intelligence technologies with an explicit indication of the places of use of the results obtained in the reports on laboratory work. Areas in which the use of artificial intelligence technologies is allowed:

- information retrieval;
- generation of initial data sets;
- development of document layouts;
- formation of alternatives when choosing.

Bonus points are awarded for:

- reports at conferences, publications on the subject of the discipline
- original approaches and algorithms to solving tasks for independent performance in laboratory work, taking into account the issues of optimization and safety.

A prerequisite for obtaining admission to the exam in the discipline is the completion of all laboratory work with the receipt of a minimum positive grade for each of them.

VIII. FUND OF ASSESSMENT TOOLS

8.1. Passport of the Appraisal Tools Fund

No p/n	Code of the direction of training, specialty	Competency Achievement Indicator Code	Name of the Valuation Tool
1	09.03.03	CPC-2.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
2		CPC-3.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
3		CPC-4.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
4		CPC-4.2	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
5		CPC-4.3.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
6		CPC-5.1.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
7		CPC-5.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
8	09.03.01	CPC-2.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
9		CPC-3.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
10		CPC-4.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
11		CPC-4.2	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
12		CPC-4.3.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
13		CPC-5.1.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
14		CPC-5.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
15		CPC-7.1.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task

No p/n	Code of the direction of training, specialty	Competency Achievement Indicator Code	Name of the Valuation Tool
16		CPC-7.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
17		CPC-7.3.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
18			– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
19			– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
20	09.03.02	CPC-2.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
21		CPC-3.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
22		CPC-4.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
23		CPC-4.2	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
24		CPC-4.3.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
25		CPC-5.1.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
26		CPC-5.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
27	09.03.04	CPC-2.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
28		CPC-3.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
29		CPC-4.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
30		CPC-4.2	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
31		CPC-4.3.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
32		CPC-5.1.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task

No p/n	Code of the direction of training, specialty	Competency Achievement Indicator Code	Name of the Valuation Tool
33		CPC-5.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
34		CPC-6.1.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
35		CPC-6.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
36	10.03.01	CPC-2.2.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
37		CPC-3.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
38		CPC-4.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
39		CPC-4.2	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
40		CPC-4.3.	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
41	09.03.01 09.03.02 09.03.03 09.03.04 10.03.01	PP-6.1 BD-3.1	– Computer Testing No1, No2 – Laboratory Work NoNo 1-5 – Exam task
42		PP-6.2 BD-3.2	– Computer Testing No2 – Laboratory Work NoNo 6-7 – Exam task

8.2. Computer Testing No1, No2

Type of assessment tool: test

List of controlled topics: Topics NoNo1-7

Form of study: full-time, part-time

1. Bank of tests by sections and topics

Section 1. Basic concepts of DB and DBMS

Part A: Choosing the correct answer

A database management system involves

- Work with multiple files while ensuring consistency
- Use of proprietary data (metadata) and knowledge that determines data integrity
- Using the Database Query Language
- synchronization of parallel access to data
- synchronization of parallel access to system files

A data dictionary is

- Table for linking data files
- Data Relationship Table
- Table to relate logical and physical data
- main memory page
- Linguistic base

The functions of direct data management in external memory are

- Provision of the necessary external memory structures for data storage
- Managing Uninterruptible Power Supplies for External Drives
- Application of algorithms for optimizing first-order predicate calculations
- Creating and Using Indexes
- Drive Integrity Verification

A transaction in a DBMS is

- A sequence of operations on the database considered by the DBMS as a whole
- a single sequence of operations on the database rejected by the DBMS
- Financial Banking Operation
- unit of user activity in relation to the database
- Delete unused data

How many root nodes are there in a hierarchical data model?

- One
- Two
- The number of root nodes is unlimited
- Three
- Five

How many attributes does a hierarchical data model node consist of?

- from only one attribute
- of one or more attributes
- from any number of attributes
- node has no attributes
- A node must not have a limit on the number of attributes

At what levels of the hierarchical data model can dependent nodes be located?

- Dependent nodes are not present in the hierarchical data model
- only on their lower
- at any level
- at all levels except the root level
- only at the first level

Which nodes is a dependent node associated with in a hierarchical data model

- with root
- with overlying nodes
- with one source
- depends on the number of attributes in the node
- with unlimited supply

How many spawned nodes can a hierarchical data model node have?

- Just one
- one or more
- depends on the number of attributes in the node
- no more than two
- Unlimited

The spawned nodes of the hierarchical data model are accessed

- via the root node
- via the source node
- through another spawned node

- via file names
- via any node

How many node instances are possible in a hierarchical data model

- no more than one
- one or more
- Unlimited
- depends on the number of attributes in the node
- Just one

The relationship between instances of nodes in a hierarchical data model is defined by

- An instance of a spawned node cannot exist without an instance of the original node
- An instance of a child node can exist without an instance of the original node
- An instance of a spawned node cannot exist without a source node
- An instance of a begotten node cannot exist with the source node
- An instance of a source node cannot exist with a spawned node

What happens when you delete an instance of a source node in a hierarchical data model

- all dependent nodes are deleted
- all instances of spawned nodes are deleted
- deletes all instances of nodes associated with the source
- No problem
- The data goes to the archive

The disadvantage of the hierarchical data model is

- Need to access only through the root node
- Difficulty in evaluating model characteristics
- difficulty in establishing connections of the 1-M type
- difficulty in establishing connections of the M-1 type
- difficulty in establishing connections of the M-M type

A dataset in a network data model is

- Named Record Set
- A set of records with common attributes
- named record
- A set of domains
- A collection of files

Number of Owner Instances in a Network Model Dataset Instance

- Any
- One
- one or more
- Two
- Three

An instance of a network model dataset has the following number of member records

- Any quantity
- One
- one or more
- An instance of a network model dataset cannot have members of
- This wording of the question is incorrect

Three levels of database presentation

- Dataology
- Conceptual
- Infological
- External
- Internal

When you delete an instance of a source node in a network data model

- The entire data set is deleted

- all instances of the source nodes are deleted
- all instances of spawned nodes are deleted
- everything is deleted
- nothing happens

A record in a network data model is involved in the following number of datasets

- in any
- in one
- In any ownership record capacity
- An entry in a network data model cannot participate in any dataset
- This wording of the question is incorrect

The disadvantage of the network data model is

- difficulties in implementing M-M type connections
- In a large number of data update anomalies
- Complexity of the model
- Difficulty in establishing 1-1 connections
- difficulties in implementing 1-M type connections

The advantages of the relational approach are

- Easily model most common domains
- the presence of a powerful mathematical apparatus
- No potential errors in the developed database
- ease of implementation of M-1 type connections
- Easy implementation of 1-M type connections

A data type in a relational data model is called

- a concept that is fully adequate to the concept of "data type" in programming languages
- Valid potential set of values of a given type
- Database Relationship Type
- Database Table Type
- Domain Type Relationship

The scheme of attitude is

- A named set of {attribute name, domain name (or type if the concept of domain is not supported)}
- Named set of pairs {attribute name, attribute size}
- Named set of pairs {attribute name, value}
- a concept fully adequate to the concept of data type" in programming languages
- Result of Boolean Expression Evaluation

A set of named relationship schemes is

- Relationship Diagram
- Tuple
- Attitude
- Database schema
- Database

A relational data model includes the following parts

- structural
- Manipulation
- Holistic
- Final
- Intermediate

In the structural part of the relational model, it is recorded that

- the only data structure used in relational databases is the normalized pair relation
- There are three approaches to the physical implementation of relationships
- There are three approaches to the physical implementation of tuples
- There are three approaches to the physical implementation of a database schema
- There are four approaches to the physical implementation of a database schema

In the manipulation part of the relational model, the following mechanisms are approved

- Relational Algebra Mechanisms
- Locking gripper mechanisms
- Information protection mechanisms
- Relational Calculus Mechanisms
- Database protection mechanisms

In the holistic part of a relational data model, the following requirements are distinguished

- Entity integrity requirements
- requirements for the description of the hardware used by the DBMS
- Integrity requirements by reference
- requirements for the description of software tools used by DBMS
- integrity requirements according to minimum criteria

A domain in a database is

- Database Relationship Type
- Allowed potential set of values of the same type
- a concept that is fully adequate to the concept of data type in programming languages
- Result of Boolean Expression Evaluation
- symbolic name in the Internet network!

The three main building blocks of the entity-relationship model

- Essence
- Recording
- Segment
- Attribute
- Communication

Serializing transactions is how they are scheduled.

- in which the cumulative effect of a mixture of transactions is equivalent to the effect of some sequential execution
- in which the net effect of the transaction mix is equivalent to the effect of some parallel execution of transactions
- in which the net effect of a mixture of transactions is equivalent to the effect of some independent execution
- Independent of the sequence of transactions
- Independent of the sequence of execution of SQL statements

Essence is

- Named data element set
- A named set of all instances of logical records of a given type
- abstraction or collective concept of an object
- abstraction or collective concept of a database
- Named set of databases

The concept of "relationship" in the "entity-relationship" model is

- A collection of physical records
- A means by which relationships between entities are represented
- Elementary Data Unit
- A complex unit of data
- A tool to support links between files

The main concepts of combining local representations include

- consistency
- Identity
- Aggregation
- Credibility
- Generalization

Aggregation when merging views is

- combining the same semantic meanings
- A data abstraction that represents different types of objects as one
- The relationship between elements is treated as a new element
- Elementary Data Unit

- A means by which relationships between entities are represented

In the modern relational DBMS, there are

- Database engine
- ND language compiler
- A set of utilities
- Hardware Suite
- A set of programming languages

Set-theoretic operations include operations

- unification of relations
- intersections of relationships
- Taking the difference in relations
- direct product of relations
- Taking the difference of tuples

Relations are compatible by association when

- have the same headings
- The headers of both relationships contain the same set of attribute names
- All headers are defined on the same domain
- attributes of the same name are defined on the same domain
- all headings are defined on the same relation

If UNION denotes a union operation, INTERSECT is an intersection operation, and MINUS is a difference operation. To denote the constraint operation, we will use the A WHERE construct, where A is the bounded relation, and sotp is a simple comparison condition. Let comp1 and sotr2 be two simple constraint conditions. Then:

- A WHERE cotr1 AND cotr2 denotes the same as (A WHERE cotr1) INTERSECT (A WHERE cotr2)
- A WHERE comp1 OR sotr2 denotes the same as (A WHERE comp1) UNION {A WHERE cotr2}
- A WHERE NOT comp1 denotes the same as A MINUS (A WHERE comp1)
- A WHERE comp1 NOT comp2 denotes the same as (A WHERE NOT comp1) UNION (A WHERE NOT comp2)
- A WHERE comp1 OR comp2 denotes the same as (A WHERE NOT comp1) UNION (A WHERE NOT comp2)

The problem with logical database design is deciding how to

- Map domain objects to abstract data model objects
- Ensure the efficiency of database queries
- Arrange data in external memory
- Create additional structures (for example, indexes)
- Arrange data in main memory

The challenge of physical database design is deciding how to

- Map domain objects to abstract data model objects
- Ensure the efficiency of database queries
- Arrange data in external memory
- Create additional structures (for example, indexes)
- Arrange data in main memory

The essence of the criterion of logical data independence is that

- The logical structure of the data is independent of the number of records stored
- The logical structure of the data is independent of the method of data storage
- The logical data structure is independent of application programs
- The logical structure of the data is independent of the computer platform
- The logical structure of the data is independent of programming languages

The essence of the criterion of physical data independence is that

- The physical location of the data does not require a change in the logical structure
- The physical location of the data does not depend on how it is stored
- The physical location of the data does not require a change in the data dictionary
- The physical structure of the data is independent of the application programs
- The physical structure of the data is independent of programming languages

The main problem with designing a relational database is making informed decisions about whether

- what relationships should the database consist of?
- What attributes should this relationship have?
- Which interface windows should represent which information groups
- how information processing should be parallelized in multiprocessor computing systems
- on which computer platform the regulatory document should be implemented

Data security is

- Protect data from unauthorized modification or destruction
- the right of individuals and organizations to transfer information to other persons and organizations
- Data protection from unauthorized access
- The logical structure of the data is independent of the method of data storage
- The logical data structure is independent of application programs

Data secrecy is

- Protection of programs from unauthorized copying or modification
- Protection of data from unauthorized modification or destruction
- the right of individuals and organizations to transfer information to other persons and organizations
- The logical data structure is independent of application programs
- The logical structure of the data is independent of the method of data storage

The key data element is

- data element from which the rest of the elements can be determined
- A data element that serves to link tables
- Data element by which the table can be identified
- Recordset
- An ordered set of records

The primary key is

- data element that allows you to find the required data file
- An item that allows you to find the required group of records
- data element that allows you to find the desired record
- a named set of data elements considered in a data processing program as a whole
- a temporary file created by a DBMS to store a single data structure

The secondary key is

- data element that allows you to find the desired record
- An item that allows you to find the required group of records
- data element that allows you to find the required data file
- a temporary file created by a DBMS to store a single data structure
- A named set of data elements considered in a data processing program as a whole

A set of values of a single data item in a relational data model is

- Domain
- primary key
- Recordset
- Secondary key
- foreign key

The creation of a two-dimensional data table and the exclusion of duplicate groups is assumed in the following normal form

- the first
- Second
- thirds
- fourth
- fifth

Separation from relations in which non-key attributes depend only on the key as a whole is done in the following normal form

- the first
- Second
- thirds

- fourth
- fifth

Allocation of attributes in which there is no transitive relationship between non-key attributes is done in the following normal form

- the first
- Second
- thirds
- fourth
- fifth

The following number of association types are used to represent one-way relationships between item pairs

- One
- Two
- Three
- four
- Five

The criteria for assessing the quality of physical methods of access to memory include

Caching

- Access efficiency
- Storage efficiency
- Total energy consumption of the machine complex
- Redundancy and integrity

Data sorting operator in SQL

- Order By
- Group By
- Sorting
- Select
- Unsorted

Expressions in Transact SQL are composed of the following elements

- External links
- Letters
- Counters
- Features
- Constants

Items have the following two properties

- redundancy and completeness
- completeness and consistency
- integrity and consistency
- redundancy and integrity
- simplicity and unambiguousness

You can get all the information from the table or only the information that meets the set conditions with the help of

- Sample queries
- Clipping requests
- queries or sorting
- change requests
- Retrieval requests

Inner is a type of union in which

- Each row in one table has many corresponding rows in another
- Each row in one table has a corresponding row in the other
- A plurality of rows in one table has a plurality of corresponding rows in another
- Each row in one table does not have a corresponding row in the other
- Each row in one table is repeated at least once

A named set of data elements, considered in a data processing program as a whole, is

- Data Syndicate
- Data aggregation
- Data Table
- Data Query
- Output of data

An attribute is

- Named Entity Characteristic
- method of data storage in DBMS
- Characteristics of associations
- Spawned Node
- Description of the location of data in the external environment

You can get rid of duplicate data when creating a query by using the

- Group By
- Where
- Having
- Distinct
- Displace

The basic concepts of a relational database include

- Attitude
- cortege'
- Domain
- Spawned Node
- Root node

The connection between the logical and physical data models is carried out through

- via attribute names
- via table names
- Through the data dictionary
- via file names
- via device names

Data integrity forms exist

- One
- Two
- Three
- four
- Five

The role of a database administrator is to

- Modifications to data files in the system
- development of database applications
- data security
- ensuring sufficient speed of DBMS operation
- Examining data for correctness

A logical description of data is

- Description of the data provided to the user
- logical level of database design
- A description of the data to determine their location
- a concept that is fully adequate to the concept of "data type" in programming languages
- Result of Boolean Expression Evaluation

The physical description of the data is

- Physical layer of database design
- a way to physically store data in an external environment
- Description of the location of data in the external environment
- physical level of database functioning
- a way to physically store data in main memory

A data file is

- a set of data aggregates
- A set of records grouped by selection criterion
- An ordered set of records
- An ordered set of tables
- An ordered set of data aggregates

An inverted file stores

- Object identifiers associated with the values of some attributes
- Object identifiers associated with the specific value of each attribute
- Object identifiers, in ascending order
- Result of Boolean Expression Evaluation
- Result of SQL Expression Evaluation

The description of the database structure is called

- user data
- data dictionary
- indices
- application data
- pages

Data redundancy in the system leads to

- Reduce the risk of data loss
- eliminating the problem of information islands
- irrational use of memory
- Data integrity issues
- Accelerate access to data

Foreign keys are used to provide

- entity integrity
- database integrity
- personal integrity
- referential integrity
- speed of data access

A supertype entity contains the following attributes

- Not general
- Unique
- General
- intact
- fractional

An entity in the ER model becomes

- Unification
- Submission
- restriction
- table
- Recording

What is the relationship of the restaurant to its 20 pizza delivery people?

- 1:M
- M:1
- M:M

Dependencies that are based on only a portion of composite primary keys are called

- transitive
- partial
- In this case, there is no attribute
- intact
- fractional

The ratio in which the minimum and maximum power is 1 is called

- mandatory
- optional
- bidirectional
- direct
- Cross-linked

What in a relationship characterizes the number of entities or participants involved

- Degree
- Power
- Strength
- Communication
- Super Key

The most common normal forms

- 1,2,3
- 4.5, Boise-Codda
- 2.4, Boise-Codda
- 2, Boisa-Codda
- 1, Boisah-Codda

A field whose values cannot be repeated can be considered

- unique
- special
- transitive
- empty
- main

What is special about the "counter" field?

- is used to enter numerical data
- is used to enter real numbers
- The data is stored elsewhere than in the field, and the field stores only a pointer to where the text is located
- has a limited size
- has the property of automatic build-up

What is the peculiarity of the "memo" field?

- is used to enter numerical data
- is used to enter real numbers
- The data is stored elsewhere than in the field, and the field stores only a pointer to where the text is located
- has a limited size
- has the property of automatic build-up

The most complete analogue of a relational database in a file model can be

- Unordered set of data
- Vector
- Family tree
- two-dimensional table
- FAT Table

How many underlying data models are there?

- One
- Two
- Three
- four
- Five

In this model, features form a directed graph with one vertex not subordinate to others

- relational
- Hierarchical
- Network

- Multidimensional
- object-oriented

The main type of communication in this model is "one with many"

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

The Basic Type of Relationship in a Hierarchical Model

- One to many
- One-to-one
- Many to many
- One to many, many to many
- No Right Answer

In which data model are objects called descendants and ancestors?

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

In which model does the transition from one record to another within the tree take place?

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

For which model is the integrity constraint: each descendant has exactly one ancestor?

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

In which data model can a descendant have any number of ancestors?

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

Data manipulation: "include in link, exclude from link" refers to the model:

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

This DBMS is easy to learn and operate and therefore is available to users with low qualifications, equipped with extensive tools for creating reports of varying degrees of complexity, created on the basis of tables of various formats.

- SQL-Server
- Visual Basic
- Visual C++
- Visual FoxPro
- Access

Which model is based on the concept of relation?

- relational
- Hierarchical
- Network
- Multidimensional
- object-oriented

An attribute that uniquely identifies an object

- primary key
- Secondary key
- Domain
- foreign key
- composite key

An attribute that serves for logical relationships between relationships

- primary key
- Secondary key
- Domain
- foreign key
- composite key

Indicate the incorrect statement

- In a relational model, relationships cannot have the same primary keys
- In a relational model, the order of strings is essential
- In a relational model, referential integrity must be respected
- In a relational model, column names must be different
- All statements are true

Specify the correct match: "The relationship key can be"

- Primary, external
- Primary, secondary
- Simple, complex
- Simple, external
- No Right Answer

In Design mode, you can

- Set data types for field names
- Delete Entries
- Sort records
- Add Entries
- No Right Answer

In Design Mode, you can't

- Work with records
- Set field names
- Set keys
- Set field properties
- No Right Answer

In Datasheet view, you can

- Work with records
- Set field names
- Set keys
- Set field properties
- No Right Answer

Indicate the incorrect statement

- The union is made over two compatible relations
- The intersection is made over two compatible relations
- The difference is made over two compatible relations
- A direct product is produced over two compatible relations
- All statements are true

Indicate the correct statement

- Two relations are compatible by union if and only if they have the same headers
- Two relations are compatible by union if and only if they have the same set of records
- Two relations are compatible by union if and only if they have the same field data type
- Two relations are compatible by union if and only if they have the same number of fields
- All statements are true

How many normal forms are there?

- One
- Two
- Three
- four
- Five

Indicate the incorrect statement, if any

- A relation is said to be reduced to 3NF if it is in 1NF and any non-key attribute is functionally fully dependent on the composite key
- A relation is said to be cast to 3NF if it is in 2NF and there are no transitive dependencies of non-key attributes on the key
- A relation is called reduced to the NBF if it is in 3NF and there are no key dependencies on non-key attributes in it
- a relation is in 4NF if it is in the NFB and there are no independent multivalued dependencies in it
- Everyone is faithful

Indicate the incorrect statement, if any

- a relation is said to be reduced to 2NF if it is in 1NF and any non-key attribute is functionally fully dependent on the composite key
- A relation is said to be reduced to 2NF if it is in 2NF and there are no transitive dependencies of non-key attributes on the key
- A relation is said to be reduced to 4NF if it is in 3NF and there are no key dependencies on non-key attributes
- a relation is in 4NF if it is in the NFB and there are no independent multivalued dependencies in it
- Everyone is faithful

Indicate the incorrect statement, if any

- A relation is said to be reduced to 2NF if it is in 1NF and any non-key attribute is functionally fully dependent on the composite key.
- A relation is called reduced to 3NF if it is in 2NF and there are no transitive dependencies of non-key attributes on the key
- A relation is called reduced to 3NF if it is in 1NF and there are no key dependencies on non-key attributes
- A relation is in 4NF if it is in the NFB and there are no independent multivalued dependencies in it.
- Everyone is faithful

Section 2. Database management systems

Indicate the incorrect statement, if any

- In Design mode, you can add entries
- In Design view, you can specify data types
- In Design mode, you can specify keys
- In Design view, you can set properties
- Everyone is faithful

Data type that is the default value in MS SQL Server

- numeric
- Symbolic
- logical
- Date
- Counter

Specify the required keywords in the structure of any SQL query SELECT statement

- SELECT, FROM
- SELECT, WHERE

- WHERE, FROM
- ORDER BY, GROUP BY
- SELECT, ORDER BY

The argument that provides the ability to eliminate duplicate values from the SELECT clause is

- ANY
- SOME
- WITH CHECK OPTION
- ALL
- DISTINCT

This aggregate function counts the number of rows or non-null field values that the query has selected

- AVG
- MIN
- COUNT
- MAX
- SUM

To order the output of table fields, SQL uses the

- ORDER BY
- HAVING
- SORT
- DISTINCT
- WHERE

Specify a special operator that always takes subqueries as arguments

- ANY
- BETWEEN
- LIKE
- IS NULL
- IN

SQL command to add information to a table

- UPDATE
- INSERT
- DELETE
- CREATE
- ALTER

Which SQL command implements the ability to modify some or all of the values in an existing table row?

- UPDATE
- INSERT
- DELETE
- CREATE
- ALTER

A sequence of SQL statements for manipulating data (read, delete, insert, modify), as a result of which there is either an impact on the database, or there is no impact at all

- CONNECTION
- UNIFICATION
- INQUIRY
- TRANSACTION
- SAMPLING

Indicate an incorrect statement

- the transaction begins with the BEGIN statement
- if the transaction is completed with a COMMIT statement, the results are committed to external memory
- when the transaction is completed with a ROLLBACK statement, the results are not in external memory
- a transaction is always terminated with a COMMIT statement
- COMMIT statement sets the commit point

A two-dimensional object consisting of rows and columns that is used to store data in a relational database.

- Table
- Performance
- Stored procedure
- Function
- Index

An attribute that specifies the type of information that can be stored in a column, parameter, or variable.

- Table
- Performance
- Data Type
- Function
- Index

A database object that can be referenced in SQL statements in the same way as a table.

- Table
- Performance
- Stored procedure
- Function
- Index

A compiled set of Transact-SQL statements that is stored under a specific name and processed as a single unit.

- Table
- Performance
- Stored procedure
- Function
- Index

A piece of code that acts as a single logical entity.

- Table
- Performance
- Stored procedure
- Function
- Index

A relational database object that provides quick access to table rows based on key values, as well as the uniqueness of rows in the table.

- Table
- Performance
- Stored procedure
- Function
- Index

A property that is assigned to a table column that prevents invalid data from being added to a column.

- Limitation
- Rule
- Silence
- Trigger
- Performance

A database object associated with columns or with user-defined data types that specifies the data values that are acceptable in a given column.

- Limitation
- Rule
- Silence
- Trigger
- Performance

A value that is automatically assigned by the system to a data, parameter, matching mode, or name when not specified by the user.

- Limitation

- Rule
- Silence
- Trigger
- Performance

A stored procedure that is executed when data is modified in a given table.

- Limitation
- Rule
- Silence
- Trigger
- Performance

CREATE TABLE

- Creates a new table in the database
- Deletes a table from the database
- Changes the structure of an existing table or integrity constraints that are set for a given table
- Creates a virtual table that corresponds to some SQL query
- Modifies a previously created view

DROP TABLE

- Creates a new table in the database
- Deletes a table from the database
- Changes the structure of an existing table or integrity constraints that are set for a given table
- Creates a virtual table that corresponds to some SQL query
- Modifies a previously created view

ALTER TABLE

- Creates a new table in the database
- Deletes a table from the database
- Changes the structure of an existing table or integrity constraints that are set for a given table
- Creates a virtual table that corresponds to some SQL query
- Modifies a previously created view

CREATE VIEW

- Creates a new table in the database
- Deletes a table from the database
- Changes the structure of an existing table or integrity constraints that are set for a given table
- Creates a virtual table that corresponds to some SQL query
- Modifies a previously created view

ALTER VIEW

- Creates a new table in the database
- Deletes a table from the database
- Changes the structure of an existing table or integrity constraints that are set for a given table
- Creates a virtual table that corresponds to some SQL query
- Modifies a previously created view

DROP VIEW

- Modifies a previously created view
- Deletes a previously created view
- Creates an index on a table to provide quick access to the attributes that are part of the index
- Drops a previously created index
- Changes the structure of an existing table or integrity constraints that are set for a given table

CREATE INDEX

- Modifies a previously created view
- Deletes a previously created view
- Creates an index on a table to provide quick access to the attributes that are part of the index
- Drops a previously created index
- Changes the structure of an existing table or integrity constraints that are set for a given table

DROP INDEX

- Modifies a previously created view
- Deletes a previously created view
- Creates an index on a table to provide quick access to the attributes that are part of the index
- Drops a previously created index
- Changes the structure of an existing table or integrity constraints that are set for a given table

DELETE

- Removes one or more rows that match the filter criteria from the base table
- Inserts data into the underlying table
- Updates the values of one or more columns in one or more rows that match the filter criteria
- Modifies a previously created view
- Creates an index on a table to provide quick access to the attributes that are part of the index

INSERT

- Removes one or more rows that match the filter criteria from the base table
- Inserts data into the underlying table
- Updates the values of one or more columns in one or more rows that match the filter criteria
- Modifies a previously created view
- Creates an index on a table to provide quick access to the attributes that are part of the index

UPDATE

- Removes one or more rows that match the filter criteria from the base table
- Inserts data into the underlying table
- Updates the values of one or more columns in one or more rows that match the filter criteria
- Modifies a previously created view
- Creates an index on a table to provide quick access to the attributes that are part of the index

COMMIT

- Complete the complex interconnected processing of information combined into a transaction
- Undo Changes Made During a Transaction
- Save the intermediate state of the database, mark it so that you can return to it in the future
- Change the set of master objects in the database, constraints that affect the entire database
- Edit a previously created retention scope

ROLLBACK

- Complete the complex interconnected processing of information combined into a transaction
- Undo Changes Made During a Transaction
- Save the intermediate state of the database, mark it so that you can return to it in the future
- Change the set of master objects in the database, constraints that affect the entire database
- Edit a previously created retention scope

SAVEPOINT

- Complete the complex interconnected processing of information combined into a transaction
- Undo Changes Made During a Transaction
- Save the intermediate state of the database, mark it so that you can return to it in the future
- Change the set of master objects in the database, constraints that affect the entire database
- Edit a previously created retention scope

ALTER DATABASE

- Complete the complex interconnected processing of information combined into a transaction
- Undo Changes Made During a Transaction
- Save the intermediate state of the database, mark it so that you can return to it in the future
- Change the set of master objects in the database, constraints that affect the entire database
- Edit a previously created retention scope

ALTER DBAREA

- Complete the complex interconnected processing of information combined into a transaction
- Undo Changes Made During a Transaction
- Save the intermediate state of the database, mark it so that you can return to it in the future
- Change the set of master objects in the database, constraints that affect the entire database

- Edit a previously created retention scope

ALTER PASSWORD

- Change password for the entire database
- Create a new database by defining the basic parameters for it
- Create a new storage area and make it available for data residency
- Delete an existing database
- Delete an existing storage area

CREATE DATABASE

- Change password for the entire database
- Create a new database by defining the basic parameters for it
- Create a new storage area and make it available for data residency
- Delete an existing database
- Delete an existing storage area

CREATE DBAREA

- Change password for the entire database
- Create a new database by defining the basic parameters for it
- Create a new storage area and make it available for data residency
- Delete an existing database
- Delete an existing storage area

DROP DATABASE

- Change password for the entire database
- Create a new database by defining the basic parameters for it
- Create a new storage area and make it available for data residency
- Delete an existing database
- Delete an existing storage area

DROP DBAREA

- Change password for the entire database
- Create a new database by defining the basic parameters for it
- Create a new storage area and make it available for data residency
- Delete an existing database
- Delete an existing storage area

GRANT

- Grant access rights for a number of actions on some database object
- Revoke access rights to an object or some actions on the object
- Defines the cursor for the query
- Open Cursor
- Count a row from the set of rows defined by the cursor

REVOKE

- Grant access rights for a number of actions on some database object
- Revoke access rights to an object or some actions on the object
- Defines the cursor for the query
- Open Cursor
- Count a row from the set of rows defined by the cursor

DECLARE CURSOR

- Grant access rights for a number of actions on some database object
- Revoke access rights to an object or some actions on the object
- Defines the cursor for the query
- Open Cursor
- Count a row from the set of rows defined by the cursor

OPEN

- Grant access rights for a number of actions on some database object
- Revoke access rights to an object or some actions on the object
- Defines the cursor for the query

- Open Cursor
- Count a row from the set of rows defined by the cursor

FETCH

- Grant access rights for a number of actions on some database object
- Revoke access rights to an object or some actions on the object
- Defines the cursor for the query
- Open Cursor
- Count a row from the set of rows defined by the cursor

PREPARE

- Execute an SQL statement that was previously prepared for dynamic execution
- Defines the cursor for the query
- Prepare an SQL statement for dynamic execution
- Create a new database by defining the basic parameters for it
- Create a new storage area and make it available for data residency

Part B: Establishing the Correct Sequence

Specify the sequence in which to obtain a relational schema from an ER schema

- The components of the entity's unique identifier are converted into the primary key
- Each simple entity turns into a table
- many-to-one (and one-to-one) relationships become foreign keys
- Indexes are created
- each attribute becomes a possible column with the same name

Build a database normalization sequence

- Second Normal Form
- Third Normal Form
- First Normal Form
- Boyce-Codd Form

Specify the keyword sequence of the correct SQL query

- WHERE
- FROM
- ORDER V U
- SELECT

Specify the sequence of adding a new record to the hierarchical database

- find the specified database tree
- insert a new record in the specified
- Jump from one entry to another inside

A typical algorithm for creating a database with data integrity support involves the following steps

- Creating primary keys
- Relationship Building
- normalization of relations
- Creating foreign keys

Set the sequence of processing the SQL query received by the DBMS

- lexical analysis
- Choosing the optimal plan
- parsing
- the product of the executed plan
- Implementation of the plan

Establish the database build logic

- Subject area
- Real-world objects
- Properties of objects and links
- Data attributes

- Values

Describe the data model - from abstract to physical

- Data model
- Infologic Data Model
- Physical data model
- Subject area
- Database

Specify the correct order of logical design of the hierarchical data model:

- Derivation of a generalized model
- Adapting the model for DBMS
- Model Optimization
- Simplifying Key Names
- Implement additional relationships

Specify the correct order of logical design of the network data model

- Derivation of a generalized model
- Adapting the model for DBMS
- Model Optimization
- Simplifying Key Names
- Implement additional relationships

Part C: Comparison of Values

Establish a correspondence (Concept - Characteristic)

- Information
- Data
- 1
 - a set of information perceived or emitted from/to the environment
 - a sequence of records placed on external storage devices and considered as a whole during processing
 - information presented in a form that allows automating its collection, storage and further processing by a person

Establish a correspondence (Concept - Characteristic)

- Primary Key
- Secondary key
 - An attribute whose value can be repeated across multiple records in a table, i.e., it is not unique
 - one or more attributes that uniquely identify the string

Establish Compliance (Normal Form - Requirement)

- First Normal Form
- Second Normal Form
- Third Normal Form
 - Relationship attributes should be simple
 - Relation attributes are simple, and each non-key attribute is functionally dependent on the key, and not transitively
 - All attributes of the relationship are simple, and every non-key attribute is functionally dependent on the key

Establish a correspondence (Concept of the method entity - relationship - actual representation)

- Essence
- Communication
- entity attribute
 - Graphically depicted association
 - real or imagined object
 - any part that serves to clarify, identify, classify, quantify or express a state

Establish Mapping (Action-Action)

- Projection
- Sampling
 - Choosing tuples
 - Selecting attributes

Specify which subsets of the SQL language these operators correspond to

- DDL Data Definition Operators
- Sauna Manipulation Operators
- DOL Data Query Language
- Controls
- Data administration
 - CREAATE TABLE
 - DELETE
 - SELECT
 - COMMIT
 - GRANT

Specify which subsets of the SQL language these operators correspond to

- DDL Data Definition Operators
- Sauna Manipulation Operators
- DOL Data Query Language
- Controls
- Data administration
 - REVOKE
 - ROLLBACK
 - SELECT
 - UPDATE
 - DROP INDEX

Specify which subsets of the SQL language these operators correspond to

- DDL Data Definition Operators
- Sauna Manipulation Operators
- DOL Data Query Language
- Controls
- Data administration
 - INSERT
 - ALTER TABLE
 - SELECT
 - GRANT
 - COMMIT

Part D: Entering the correct answer

Specify the occurrence predicate of the SELECT statement

Specify a predicate for specifying a range of values for the SELECT operator

Specify a comparison predicate with a sample SELECT operator

Specify a comparison predicate with an undefined value of the SELECT operator

Specify a SELECT statement keyword that excludes duplicate rows from the result set

Specify an aggregate function that defines the number of rows or non-empty field values that the query has selected

Specify an aggregate function that defines the sum of all selected values for a given field

Specify an aggregate function that defines the arithmetic mean of all selected values for a given field

Specify the aggregate function that defines the smallest of all the selected values for this field

Specify the aggregate function that defines the highest of all the selected values for the field

Specify the SQL table creation statement

Specify the SQL Table Delete Operator

Specify the SQL Language Representation Creation Statement

Specify the SQL Language Representation Delete Operator

Specify the SQL index creation operator

Specify the SQL Index Delete Statement

Specify a statement to delete a row from the underlying SQL table

Specify the operator to insert a row into the underlying SQL table

Specify a row update operator for the underlying SQL table

Specify the SQL Transaction Completion Statement

2. Test specification

Purpose of the test. The test can be used to conduct a milestone (with a partial sample of test tasks) and final control of the level of knowledge of 2nd year students.

Elements of the content of the course "Databases and DBMS" included in the test.

Basic concepts of DB and DBMS.

Purpose and main components of the database system. Database presentation layers. An overview of modern database management systems.

Database design. Database architecture. Data models. Hierarchical, network, relational data models. The "entity-relationship" model. Design levels: conceptual, logical, physical. Database design based on normalization principles.

Mathematical Foundations of Relational Data Manipulation. Single-schematic and multi-schematic relations. Basic operations of relational algebra. Traditional and specialized surgeries.

SQL language. Relational data model. Operations on relationships. Relational algebra. The structure of the SQL language. DDL data definition operators. DDL data manipulation operators. DQL query language. Data types. The SELECT statement. Queries: simple, join-based, nested queries.

Distributed data processing. Remote data access model. Parallel processes. Transaction model. Transaction properties. Problems of parallel processes. Transaction conflicts and how to resolve them. Database security.

Database management systems.

MS SQL Server architecture. Installation and configuration of MS SQL Server. Basic concepts of the Transact-SQL language.

Work with MS SQL Server databases.

Work with MS SQL Server tables.

MS SQL Server data sampling. Adding, modifying, and deleting data.

Indexing. Views. Cursors.

MS SQL Server transactions.

Stored procedures, user-defined functions, MS SQL Server triggers.

Administration of MS SQL Server.

List of objects of control (**types of knowledge, skills controlled by test tasks**).

1	Definition of database design goals, efficiency criteria, constraints.
2	System analysis of database design, subject area and their interconnections.
3	Selection of input data for database design.
4	Development of generalized options for solving the problem of database design, analysis of the considered options, forecasting the consequences, finding compromise solutions in conditions of multi-criteria and uncertainty, planning the implementation of databases.
5	Assessment of the reliability and quality of database functioning.
6	Database development technology, database testing and debugging.

7	Creation and research of mathematical and program models and methods used in the creation of databases.
8	Development and improvement of formal models and methods used in database design.
9	Organization of individual stages of the database development process with a given quality and in a given time.
10	Selection of technology, tools and computer tools when organizing the process of database development.
11	Evaluation, control and management of the database development process.
12	Installation, configuration and maintenance of system and application database software.

Distribution of test work tasks by level of complexity.

The first part of the work (A 1 – A 160) checks the assimilation of educational material by students at the basic level of complexity.

All tasks of the second part of the work (B 1 - B 10) and the third part (C 1 - C 10) belong to an increased level of complexity.

Tasks D 1 – D 20 is a task of a high level of difficulty.

Difficulty level of tasks	Number of tasks	Maximum primary score	Percentage of the maximum primary score for tasks of this level of complexity from the maximum primary score for the entire work, equal to 260 points
Basic	160	160	62 %
Increased	20	40	15 %
High	20	60	23 %
Total	200	260	100%

The versions of the exam test are equivalent in difficulty, the same in structure, parallel in the arrangement of tasks: under the same number in all versions of the work there are tasks that check the same element of content.

The level of difficulty is determined by:

- the content of the task;
- the number of actions that need to be performed to complete the task;
- variability of these actions.

3. The structure of the test by the forms of test tasks. Examples of instructions for tasks.

The total number of tasks in the bank of test questions is 200.

The test consists of four parts. For each module of the discipline, 20 questions corresponding to the studied topics are randomly selected from the bank of test questions. The distribution of test questions is carried out in the following proportion: 10 questions from Part A, 5 questions from Part B, 3 questions from Part C, 2 questions from Part D. Points for the correct answer are distributed evenly among the questions (0.5 points for the correct answer), without taking into account the complexity of the question.

PART A. Tasks No A 1 – A 160 are closed text tasks that require choosing the correct answer from several proposed ones.

INSTRUCTIONS FOR THE TASKS OF PART A OF THE TEST.

Choosing the correct answer. Indicate the number of the correct answer (if there are several correct answers, then they should be written as one number, without separating signs).

PART C. Tasks No B 1 – B 10 are closed test tasks to establish the correct sequence.

INSTRUCTIONS FOR THE TASKS OF PART B OF THE TEST.

Establishing the correct sequence. Specify the sequence of numbers of the correct answer options (the numbers of answers are written as one number, without separating characters).

PART C.

Tasks C 1 – C 10 are closed test tasks for comparing values.

INSTRUCTIONS FOR THE TASKS OF PART FROM THE TEST.

Establishing compliance. Establish the correct correspondence between the task in the narrative and the answer in the definition area.

PART D.

Tasks D 1 – D 20 are open-ended test tasks that require the correct answer to be entered.

INSTRUCTIONS FOR THE TASKS OF PART D OF THE TEST.

Enter the correct answer.

4. Evaluation criteria:

The test is given to students in modules 1 and 2. For each module, 20 questions are selected from the general question bank that correspond to the topics studied. For each module of the discipline, 20 questions corresponding to the studied topics are randomly selected from the bank of test questions. The distribution of test questions is carried out in the following proportion: 10 questions from Part A, 5 questions from Part B, 3 questions from Part C, 2 questions from Part D. Points for the correct answer are distributed evenly among the questions (0.5 points for the correct answer), without taking into account the complexity of the question.

For passing the test based on the results of studying the module, the student receives:

- **10-9 points in case of correct, complete and accurate answer to 85-100% of questions;**
- **7-8 points** in case of correct, complete and accurate answers to 71-84% of questions;
- **5-6 points** in case of correct and accurate answers to 60-70% of questions
- **less than 5 points-** if you have answered 59% or less of the proposed questions

8.3. Laboratory work No1

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No3

Form of study: full-time, part-time

- 1) It is necessary to select and approve the version of the task from the file with the teacher.
- 2) Describe the subject area extensively and design your database for the selected subject area
- 3) Create an ER diagram based on an extended domain description
- 4) Based on the obtained ER diagram, get a DDL script for database generation;

Evaluation criteria (full-time):

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

Evaluation criteria (part-time):

7 points are given to a student if he or she has completed all the tasks provided for in the laboratory work in a timely manner, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

6 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

5 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.4. Laboratory Work No2

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No4

Form of study: full-time, part-time

Assignment for laboratory work:

1. To implement at the physical level a personal database designed in accordance with the variant of the task in the first laboratory work.
2. Configure restrictions (at least one of each type): NULL, NOT NULL, DEFAULT, UNIQUE, CHECK for database tables.
3. Check all connections and configure them for data integrity.
4. Fill in the database data, at least 10 records for each table.
5. Check the performance of the database (including the correctness of restrictions) by implementing at least 5 queries of each type for each table: SELECT, INSERT, UPDATE, DELETE.
6. Create indexes for foreign keys on child tables and for data lookups.
7. Create a script to restore the database.

Contents of the report:

1. Database creation script
2. Scripts for commands to create/modify table constraints
3. Data insertion scripts and screenshots of the results of their work
4. Data sampling scripts and screenshots of their results
5. Scripts for changing data and screenshots of the results of their work
6. Data deletion scripts and screenshots of the results of their work

Evaluation criteria (full-time):

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the

academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

Evaluation criteria (part-time):

7 points are given to a student if he or she has completed all the tasks provided for in the laboratory work in a timely manner, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

6 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

5 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.5. Laboratory Work No3

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No5

Form of study: full-time, part-time

Tasks of the laboratory laboratory

1. For each table, write a query for a selection of fields and records, ordering them according to some criterion for displaying information from your subject area, using ORDER BY.

2. According to the logic of the subject area, write 2 queries to tables with a selection of several records according to a certain condition, using WHERE and logical operators.

3. According to the logic of the subject area, write 2 queries to the tables: with a selection of the first N records and M records according to a certain condition, using WHERE and ORDER BY.

4. Create 1 query that outputs data from three or two fields without duplicating records.

5. According to the logic of the subject area, write 4 queries to tables with a selection of records according to a certain condition using:

a. a special IN operator using WHERE and Boolean operators;

b. a special BETWEEN operator using WHERE and ORDER BY;

c. a special LIKE operator using all possible templates;

d. the special operator EXISTS.

6. According to the logic of the subject area, write 3 queries using: UNION [ALL] , EXCEPT, INTERSECT.

7. According to the logic of the subject area, write 1 query using two or three operators UNION [ALL] , EXCEPT, INTERSECT in one query at once.

8. According to the logic of the subject area, create 5 views to tables with a selection of records from several tables according to a certain condition using:

- a. INNER JOIN
- b. LEFT JOIN
- c. RIGHT JOIN
- d. FULL JOIN
- e. CROSS JOIN

9. Create a materialized representation for obtaining statistical information based on aggregation functions with one field and based on the grouping of several fields.

For each query, you must provide a formal description of the query (query logic), its corresponding SQL statement, the source data in the tables, and the results of the query.

The report (in PDF format) should contain a full description of all created requests, in the form of the following information:

- Query type, for example: "Selection of information with special operators in the predicate";
- Logical formulation of the request, for example: "Display all information about buyers who bought real estate in 2012";
- Implementation of the query on a SQL script;
- Initial data for the script;
- The result of the output or the script.

Evaluation criteria (full-time):

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

Evaluation criteria (part-time):

7 points are given to a student if he or she has completed all the tasks provided for in the laboratory work in a timely manner, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

6 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the

academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

5 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.6. Laboratory Work No4

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No6

Form of study: full-time, part-time

Assignment for laboratory work. Implementation of business logic inside the database

1. Determine the time of year for a given date: autumn, winter, spring or summer.

2. The user's date of birth is known. Check whether he is 16 years old or not.

3. Set whether the string of characters is an email address.

4. Using the Caesar cipher, encrypt the given line of text. The idea of this method of encryption is that the alphabet is placed as if clockwise. For encryption, the letters of the text are replaced by letters separated by a specified number of letters (shift) clockwise.

5. The user's surname, first name and patronymic are known. Find the number of his personality. The rule for obtaining a personality number: each letter is associated with a number - the ordinal number of a letter in the alphabet. These numbers are added, if the resulting sum is not a single digit, then the digits of the number are added up again and so on until a single number is obtained.

Features:

1. Create a scalar custom function that calculates the geometric mean of two numbers or dates based on the data in your database.

2. Create a scalar user function to define the day of the week of the entered date.

3. Create a tabular user function designed to combine several text fields of a table into a single output field displayed in uppercase.

4. Create a table function that collects all the data about the user from your table: Last name, Phone + email, Rest of the information, if something is missing, then add the word "No", e.g., "Phone No"

5. Write scalar and table functions that implement the business logic of your database.

Triggers:

1. Create a trigger that displays the message "Record added" when adding a record to the table.

2. Create a trigger that saves all deleted records to the table log with additional information about who deleted and when, when deleting records from the table.

3. Create a trigger that provides cascading deletion of related records from multiple tables.

4. Create a trigger that implements business logic for your database.

Evaluation criteria (full-time):

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the

academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

Evaluation criteria (part-time):

7 points are given to a student if he or she has completed all the tasks provided for in the laboratory work in a timely manner, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

6 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

5 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.7. Laboratory Work No5

Control type: ongoing

List of controlled topics (modules): Topic No7

Form of study: full-time, part-time

Task 1

Run an experiment: with transactions

Create a simple table that contains three fields – a surrogate key and two attributes that contain rows.

Use the script to populate the table with data.

1.1. Use the EXPLAIN ANALYZE functions to measure the execution time of a query that displays the contents of the table sorted by column CODE_1.

1.2. Add an index to column CODE_1.

1.3. Repeat the previous two operations.

1.4. Compare the time obtained. How many times has it changed? Enter the results of the calculation in the table.

Task 2

2.1. Make a query to the Test table that displays all rows in sorted order in which column CODE_1 begins with the 'a' symbol.

2.2. Analyze the received request and explain the result.

Task 3

Analyze your personal database – you created views and functions in a previous lab.

Select the 2 most difficult queries and evaluate their performance using the EXPLAIN ANALYZE command.

3.1. Next, build an index to speed up a single query. Explain your choice using the EXPLAIN ANALYZE command.

3.2. Build a covering index and explain your choice using the EXPLAIN ANALYZE command.

Task 4

Suppose a teacher goes to an electronic journal to give a student a grade for a discipline. At the same time, the student decided to check whether he had been given a grade.

4.1. Simulate this situation by capturing the actions of the student and the teacher in a transaction with different isolation levels.

4.2. Describe the work in different sessions with different isolation levels.

This simulation can be adapted based on your database.

Task 5

Select a text field in your database, fill it in with large lines. Create a copy of this field for full-text search. (Lecture No12)

5.1. Construct 2-3 selection queries by full-text field.

5.2. Evaluate them using the EXPLAIN ANALYZE command.

5.2. Build a GIN index on this field and evaluate previous queries to use this index again using the EXPLAIN ANALYZE command.

Contents of the report.

For each query, you must provide a formal description of the query (query logic), its corresponding SQL statement, the source data in the tables, and the results of the query.

The report (in PDF format) should contain a full description of all created requests, in the form of the following information:

Implementation of the query on a SQL script;

- Initial data for the script;
- The result of the output or operation of the script.

Evaluation criteria (full-time):

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

Evaluation criteria (part-time):

7 points are given to a student if he or she has completed all the tasks provided for in the laboratory work in a timely manner, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

6 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the

academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

5 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.8. Laboratory Work No6

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No7

Form of study: full-time, part-time

It is necessary to study and apply the work on your database:

- with arrays
- with JSONB
- with data encryption

The work is carried out in accordance with the methodological instructions issued by the teacher.

Evaluation criteria:

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.9. Laboratory Work No7

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No8

Form of study: full-time

Goal: to master the work with document-oriented and search DBMS.

Objectives:

1. Create a document store (MongoDB or Elasticsearch).
2. Upload a collection of texts (e.g., news, reviews, articles).
3. Implement indexing and content search.
4. Connect a Python script for text analysis (e.g., keyword extraction, classification).
5. Evaluate storage and retrieval efficiency.

Evaluation criteria:

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.10. Laboratory Work No8

Valuation Tool Type: Report Protection

List of controlled topics (modules): Topic No8

Form of study: full-time

Objective: To use vector databases to implement semantic searches.

Objectives:

1. Get embeddings of texts or images (using Sentence Transformers, CLIP, etc.).
2. Deploy on-premises with FAISS or Milvus.
3. Save embeddings and implement nearest neighbor search.
4. Integrate search into a simple application (for example, a chatbot with RAG).
5. Compare search results by keywords and by meaning.

Evaluation criteria:

5 points are given to a student if he/she has timely completed all the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. He was able to answer additional questions related not only to the process of performing laboratory work, but also to the understanding of the actions performed and the tasks solved.

4 points are given to a student if he/she has completed 80% of the tasks provided for in the laboratory work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of theoretical knowledge in the amount of the content of the academic discipline related to laboratory work. Was able to answer questions related to the process of performing laboratory work.

3 points are given to a student if he/she has completed more than 50% of the tasks set in the laboratory work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

0 points are given to a student if he/she has not completed the tasks set in the laboratory work, is not able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to laboratory work.

8.11. Exam task

Structure of the exam task

The discipline of Database and DBMS involves passing the exam in a practice-oriented format. The exam task includes two sections:

1. Control of theoretical knowledge.

This type of control is carried out automatically by conducting computer testing on the basis of the question bank specified in clause 10.2.

To pass the test, the student is allocated 30 minutes, during which an individual bank of 20 test questions is formed by means of the ICTIB (lms.sfedu.ru) e-learning system. The distribution of test questions is carried out in the following proportion: 10 questions from Part A, 5 questions from Part B, 3 questions from Part C, 2 questions from Part D. Points for the correct answer are distributed evenly among the questions (0.5 points for the correct answer), without taking into account the complexity of the question.

For passing the test based on the results of studying the module, the student receives:

- 20-17 points in case of correct, complete and accurate answer to 85-100% of questions;
- 13-16 points in case of correct, complete and accurate answers to 71-84% of questions;
- 10-12 points in case of correct and accurate answers to 60-70% of questions
- less than 10 points - if he answered 59 or less % of the proposed questions

2. Control of practical knowledge.

This type of control is carried out by automated verification of the results of the development/modification of the database structure and writing scripts for working with it.

The task for testing practical knowledge is developed individually for each examinee and includes the following thematic components:

1. Create database tables.
2. Modification of database tables.
3. Create procedures, functions, indexes, triggers.
4. Selecting data and entering data into the database.
5. Database administration

The practical task includes 10 items, the weight of each of which is from 1 to 4 points. Points for the practical part are formed by summing up the points for successfully completed items. Partially completed points or points completed with errors are evaluated as 0 points.

The final grade is determined by summing up the points earned by the student when completing the practical task and answering theoretical questions. If the sum of the points scored by the student is less than 22, then 0 points are given for the exam (unsatisfactory).